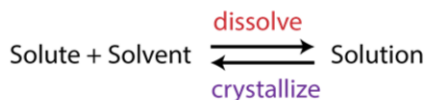
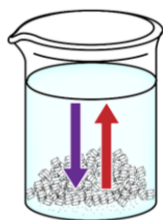


The solution process involves both dissolution and crystallization of solute



Dynamic equilibrium
rate of dissolution = rate of crystallization

Solubility: The maximum amount (in g or mol) of solute that can dissolve in a volume of solvent under a given set of conditions (T and P).

Factors affecting solubility:

- Type of IMFs and their strengths
- Temperature
- Pressure

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We have already seen that dissolution can occur when a solid solute is combined with a solvent. The reverse of this process, the precipitation of a crystalline solid from a solution, can occur as well.

Imagine a scenario in which a very large amount of salt is added to a small quantity of solvent. Initially the rate of dissolution will be much faster than the rate of crystallization since the solvent initially contains no solute that can crystallize. As more solute dissolves, the concentration of solute in the solution increases, thus increasing the rate of crystallization as well. Eventually the solute concentration will reach a point where the rate of crystallization equals the rate of dissolution (i.e., a state of dynamic equilibrium). At that point, there will no longer be anymore net dissolution of the solute. This maximum solute concentration, at which the rates of dissolution and crystallization are equal, is known as the **solubility** of the solute.

The solubility of a solute is the maximum amount in grams or moles of a solute that can dissolve in a specific volume of solvent under a specific set of conditions. A solute's solubility is influenced by the types of IMFs the solute forms with itself and with the solvent. Solubility is also affected by temperature and can increase or decrease with temperature depending on the salt and solvent.. The solubility of gases is affected by pressure as well.

Saturated solutions

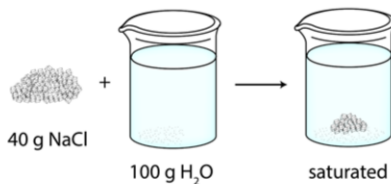
Unsaturated

A solution that contains less solute than is soluble under those conditions (no undissolved solute).



Saturated

A solution that contains the maximum amount of solute than is soluble under those conditions (undissolved solute is present).



Supersaturated

A solution that contains MORE than the maximum amount of solute than is soluble under those conditions (solution is unstable).

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The terms unsaturated, saturated, and supersaturated are used to describe how much solute is dissolved in a solution relative to the solubility of the solute under those conditions.

Unsaturated solutions contain less solute than is soluble under a given set of conditions. For example, the solubility of NaCl in water is 36 g per 100 g of water at room temperature. Thus, 30 g of NaCl will dissolve completely in 100 g of water. The resulting solution will contain less solute than is soluble under those conditions (30 g versus 36 g), meaning that the solution is unsaturated.

Saturated solutions contain the maximum amount of solute that is soluble under a given set of conditions. For example, if 40 g of NaCl is combined with 100 g of water at room temperature, only 36 g of the NaCl will dissolve since the solubility of NaCl in water is 36 g per 100 g of water this temperature. The remaining 4 g of NaCl will remain as a solid at the bottom of the solution. The resulting solution is said to be saturated since it contains the maximum amount of solute that is soluble under those conditions (36 g).

Supersaturated solutions contain *more* dissolved solute than is soluble under those conditions. These solutions are unstable, and given the opportunity, the excess solute will precipitate from the solution. Supersaturated solutions are typically created by changes in pressure or temperature that lower the solubility of a solute, such as cooling the sample. For example, at 100°C 39 g of NaCl can dissolve in 100 g of water. If a saturated solution of NaCl is prepared at 100°C and then cooled to room temperature (where NaCl is less solubility), the solution will be supersaturated, at least until some of the dissolved solute precipitates from the solution.